

Lesson 06 Practice: CPU vs GPU & Microprocessor

JKSSB Computer Awareness

October 10, 2026

Provenance: All 20 items are original JKSSB-pattern questions written for this lesson. None is a real PYQ; each is labelled *[Original]*.

Q1. *[Original]* Which statement best describes the cores of a CPU?

- A. Many simple cores optimised for throughput
- B. A single storage unit
- C. Few powerful general-purpose cores optimised for latency
- D. No cores at all

Q2. *[Original]* A GPU is designed primarily for:

- A. sequential general-purpose work only
- B. parallel throughput tasks such as graphics and pixels
- C. power management
- D. secondary storage

Q3. *[Original]* Serial processing means that:

- A. multiple tasks run simultaneously
- B. one task is done at a time, in sequence
- C. work is split across many cores
- D. no instructions are executed

Q4. *[Original]* Parallel processing means that:

- A. one task is done at a time in strict order
- B. only graphics work is performed
- C. multiple tasks or operations proceed simultaneously
- D. a GPU is always required

Q5. *[Original]* Which statement about GPU cores is correct?

- A. GPU cores are identical to general-purpose CPU cores.
- B. GPU cores replace the CPU entirely.
- C. GPU cores are used only for storing files.
- D. GPU cores are simpler and more numerous, optimised for parallel throughput.

Q6. *[Original]* GPU stands for:

- A. General Processing Unit
- B. Graphics Processing Unit
- C. Graphical Parallel Unit
- D. Global Processing Utility

Q7. *[Original]* CPU stands for:

- A. Central Processing Unit
- B. Computer Peripheral Unit
- C. Central Program Utility
- D. Core Processing Unit

Q8. *[Original]* A microprocessor is best described as:

- A. a type of secondary storage
- B. a CPU implemented on a single chip
- C. a complete computer with memory and I/O on one chip
- D. an output device

Q9. *[Original]* A microcontroller is best described as:

- A. a CPU on a single chip with no memory or I/O
- B. a large mainframe processor
- C. a GPU with many cores
- D. a microprocessor plus integrated memory and I/O peripherals on one chip

Q10. *[Original]* Microcontrollers are typically used in:

- A. embedded and special-purpose systems
- B. supercomputers only
- C. plotters and printers only
- D. cloud data centres only

Q11. *[Original]* Hyper-threading allows:

- A. the physical core count to double
- B. one physical core to present two logical threads
- C. the clock speed to double
- D. multitasking to be disabled

Q12. *[Original]* Evaluate the statements:

- I. A CPU typically has fewer, more powerful cores than a GPU.
- II. A GPU has many simple cores designed for parallel throughput.
- III. GPU cores are fully general-purpose replacements for CPU cores.

- A. I only
- B. I and II only
- C. II and III only
- D. I, II, and III

Q13. *[Original]* L1 cache is:

- A. the largest and slowest cache level
- B. the smallest and fastest level, closest to the core
- C. stored on the hard disk
- D. the same as main memory

Q14. *[Original]* Which order lists cache from smallest and fastest to largest and slowest?

- A. L3, L2, L1
- B. L2, L1, L3
- C. L1, L2, L3
- D. L1, L3, L2

Q15. *[Original]* Clock speed is typically measured in GHz and expresses:

- A. storage capacity
- B. memory size
- C. cycles per second
- D. number of cores

Q16. *[Original]* A multi-core processor is one that has:

- A. multiple independent processing units (cores) on one chip
- B. multiple monitors
- C. additional RAM
- D. a single slower core

Q17. *[Original]* An office desktop has a quad-core 2.4 GHz CPU, 16 GB RAM and an SSD, yet one particular program runs slowly. What is the most appropriate diagnosis?

- A. The CPU must be faulty, because more cores always means faster.
- B. The GPU must be missing.
- C. The clock speed must be zero.
- D. A single-threaded program or a memory/disk-I/O bottleneck — not automatically a “slow CPU.”

Q18. *[Original]* Which statement is **NOT** correct?

- A. A CPU has few, powerful general-purpose cores.
- B. A GPU has many simpler cores for parallel work.

- C. A microprocessor integrates a CPU on a single chip.
- D. A microcontroller is just a CPU with no memory or I/O.

Q19. *[Original]* The key difference between a microprocessor and a microcontroller is:

- A. clock speed only
- B. colour and size only
- C. the level of integration — the microcontroller includes memory and I/O on the same chip
- D. the microprocessor is analog

Q20. *[Original]* Which workload is best suited to a GPU?

- A. a single sequential spreadsheet calculation
- B. storing files
- C. printing a document
- D. processing millions of pixels or parallel floating-point operations

Q21. *[JKSSB FAA-pattern]* The Intel 4004 is historically important as the first:

- A. commercial microprocessor.
- B. hard disk drive.
- C. programming language.
- D. optical mouse.

Q22. *[JKPSI-pattern]* A GPU is best suited for which kind of workload?

- A. Massively parallel graphics and numeric tasks
- B. Power supply regulation
- C. Paper printing control
- D. Keyboard input scanning

Q23. *[SSC CGL-pattern]* Which company developed the Pentium series of processors?

- A. Intel
- B. A paper mill
- C. A textile firm
- D. A shipping company

Q24. *[JKSSB-pattern]* (**MSQ: select ALL correct options.**) Which of the following are associated with microprocessors?

- A. Intel
- B. AMD
- C. ARM-architecture designers
- D. Printer ink cartridges

- Q25.** *[SSC CGL-pattern]* (MSQ: select ALL correct options.) Which statements about processors are correct?
- A. A microprocessor places the CPU on a single chip.
 - B. A GPU excels at parallel data-heavy tasks.
 - C. Processor clock speed is measured in hertz.
 - D. RAM chips sit inside the microprocessor package by definition.
- Q26.** *[Original]* Which processor is generally optimized for a wide range of sequential general-purpose tasks?
- A. CPU
 - B. GPU
 - C. Network interface card
 - D. Sound card
- Q27.** *[Original]* A graphics workload applies the same operation to many pixels in parallel. Which component is commonly suited to this work?
- A. GPU
 - B. Hard disk
 - C. Keyboard controller
 - D. ROM
- Q28.** *[Original]* A microcontroller is most likely to combine a CPU with:
- A. Only an external display
 - B. Memory and input/output peripherals on one chip
 - C. A separate supercomputer
 - D. A printer interface only
- Q29.** *[Original]* Which is a typical example of an embedded-system use for a microcontroller?
- A. Dedicated washing-machine control
 - B. A large-scale weather simulation
 - C. A multi-user banking database server
 - D. A general-purpose desktop operating system
- Q30.** *[Original, MSQ drill]* (MSQ: select ALL correct options.) Which statements are generally correct?
- A. CPUs are designed for broad general-purpose processing.
 - B. GPUs can execute many similar operations in parallel.
 - C. Microcontrollers commonly serve embedded control tasks.
 - D. A GPU is another name for a hard disk.